
Semantic Search Test Dataset (Sample)

You can copy this into a doc and export as PDF.

◆ Pair 1 (Geospatial – Strongly Related)

BLOCK A1

The Haversine formula is used to calculate the shortest distance between two points on the Earth's surface using their latitude and longitude. It is widely applied in navigation systems and mapping applications.

BLOCK A2

Radius-based geospatial queries rely on accurate distance calculations between coordinates. Many systems use formulas like the Haversine method to determine which locations fall within a specified range.

✓ Expected: A1 ↔ A2 should match strongly

◆ Pair 2 (Databases – Conceptual Link)

BLOCK B1

Indexing in databases improves query performance by organizing data in a way that allows faster retrieval. Common indexing structures include B-trees and hash indexes.

BLOCK B2

Efficient data retrieval systems rely on indexing techniques to minimize search time. Structures like B-trees enable databases to quickly locate records without scanning entire tables.

✓ Expected: Strong similarity

◆ Pair 3 (Machine Learning – Paraphrase)

BLOCK C1

Overfitting occurs when a machine learning model learns noise in the training data instead of the underlying pattern, reducing its performance on unseen data.

BLOCK C2

A model that memorizes training data rather than generalizing patterns is said to overfit. This leads to poor accuracy when evaluated on new datasets.

✓ Expected: Very strong similarity

◆ Pair 4 (Networking – Indirect Relation)

BLOCK D1

Latency in a network refers to the time it takes for data to travel from the source to the destination. High latency can negatively impact real-time applications.

BLOCK D2

Real-time systems such as video calls and online gaming require low response times to function effectively. Reducing delays in data transmission is critical for performance.

✓ Expected: Medium–strong similarity

◆ Pair 5 (Your Original Problem – Improved Version)

BLOCK E1

Distance calculation is a fundamental part of geospatial systems. Algorithms like the Haversine formula are used to compute distances between coordinates in mapping and navigation applications.

BLOCK E2

Geospatial search systems perform radius queries to find nearby locations. These systems depend on accurate distance computation between coordinates to return relevant results.

✓ Expected: Strong similarity (this fixes your earlier issue)

◆ Pair 6 (Negative Example – Should NOT Match)

BLOCK F1

Photosynthesis is the process by which plants convert sunlight into energy using chlorophyll.

BLOCK F2

Blockchain is a distributed ledger technology used to securely record transactions across multiple systems.

✗ Expected: Low similarity
